# Department of Computer Science & Engineering, DIAT

Deep Learning (CE631)
MTech/PhD

**Assessment 1 (23/02/2023)**
**Maximum Marks: 10**

Q1. Identify the significance of 'e' during the calculation of softmax activation [1]

a) To get output value between 0 and 1
b) To map output on real numbers
c) To avoid negative values
d) None of the above

Q2. If we use very high value for w in logistic function, then it resembles as................? [1]

Q3. Define any 02 activation functions used in deep learning with graphical representations. Is it necessary to use differentiable activation functions ? Justify. [2]

Q4. How can Bias and Variance in the Error is associated with Overfitting and Underfitting. Explain diagramaticatlly. [2]

Q5. Consider a discrete random variable **X** which can take one of 3 values from the set {x1, x2, x3}. A distribution over X defines the value of Pr(X = x) $\forall x \in$ {x1, x2, x3}. Consider two such distributions **P** and **Q** which are defined as follows, What will be the KL divergence between P and Q? (Use base-2 logarithm.) [2]

| Probability Distribution | x1 | x2 | x3 |
|---|---|---|---|
| P(X) | 0 | 1 | 0 |
| Q(X) | 0.25 | 0.35 | 0.40 |

Q6. Differentiate any two Gradient Descent Methods with example. [2]

# Department of Computer Science & Engineering, DIAT

Deep Learning (CE631)
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**Assessment 2 (22/03/2023)**
**Maximum Marks: 10**

**Q1.** Given a 128x128x3 image and 6 filters of size 9x9x3, what will be the dimension of the output volume when the stride S=1 and the padding P=2? [1]

Q2. Does the size of the feature map always reduce upon applying the filters? Explain why or why not. [2]

Q3. Explain the significance of the RELU Activation function in Convolution Neural Network. [2]

Q4. Consider the Code Snippet shown in Fig. 1 and calculate the output of model.summary(). Here, we have no Non-Trainable Parameters. [5]

```
model = models.Sequential()

model.add(layers.Conv2D(32, (5,5), activation='relu', input_shape=(28, 28,1)))
model.add(layers.MaxPooling2D((2, 2)))

model.add(layers.Conv2D(64, (5, 5), activation='relu'))
model.add(layers.MaxPooling2D((2, 2)))

model.add(layers.Flatten())
model.add(layers.Dense(10, activation='softmax'))

model.summary()
```

Figure 1. Code Snippet

# Department of Computer Science & Engineering, DIAT

<u>Deep Learning (CE631)</u>
MTech/PhD

**<u>Assessment 3 (01/05/2023)</u>**
**<u>Maximum Marks: 10</u>**

Q1. Can you think of a few applications for a sequence-to-sequence RNN? What about a seuence-to-vector RNN? And a vector-to-sequence RNN? [2.5]

Q2. Why do people use encoder–decoder RNN rather than plain sequence-to-sequence RNNs for automatic translation? [2.5]

Q3. How could you combine a convolutional neural network with an RNN to classify videos? [2.5]

Q4. Suppose you want to train a classifier and you have plenty of unlabeled training data, but only a few thousand labeled instances. How can autoencoders help? How would you proceed? [2.5]

**DEFENCE INSTITUTE OF ADVENCED TECHNOLOGY**

**(Deemed University)**

MTech/MS/PhD EXAMINATIONS

Quiz-I EXAMINATION – (SPRING/~~AUTUMN~~ 2023)

Course (Subject): Computer Vision

Code: CE-632

Duration : 01 HOURS

Max Marks: 10

Date: 21/02/2022 Time: 12.00-13.00Hrs

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Note:

1. Answer **ALL** questions.
2. Marks are indicated against each Question.
3. Question paper consists of: 01 Pages.

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**Q1.** **[5 Marks]**

Explain any Image Thresholding technique in detail.

**Q2.**

Write a short note on connected component labelling algorithm. **[5 Marks]**

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**DEFENCE INSTITUTE OF ADVENCED TECHNOLOGY**

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MTech/MS/PhD EXAMINATIONS

Quiz-II EXAMINATION – (SPRING/~~AUTUMN~~ 2023)

Course (Subject): Computer Vision

Code: CE-632

Duration : 01 HOURS

Max Marks: 10

Date: 21/03/2022 Time: 12 Hrs.-13 Hrs.

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Note:

1. Answer **ALL** questions.
2. Marks are indicated against each Question.
3. Question paper consists of: 01 Pages.

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**Q1.** **[5 Marks]**

Explain working of Canny Edge Detection Algorithm.

**Q2.** **[5 Marks]**

Explain Hough Transform for circle detection in detail.

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# DEFENCE INSTITUTE OF ADVENCED TECHNOLOGY

## (Deemed University)

MTech/MS/PhD EXAMINATIONS

Quiz-III EXAMINATION – (SPRING/~~AUTUMN~~ 2023)

Course (Subject): Computer Vision | Code: CE-632

Duration : 01 HOURS | Max Marks: 10

Date: 25/04/2022 Time: 12 Hrs.-13 Hrs.

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Note:

1. Answer **ALL** questions.
2. Marks are indicated against each Question.
3. Question paper consists of: 01 Pages.

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**Q1.**

Explain working of any Corner Detector. **[5 Marks]**

**Q2.**

Explain Boundary tracking procedure in detail. **[5 Marks]**

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**(Deemed University)**

MTech/MS/PhD EXAMINATIONS

END SEMESTER EXAMINATION – (SPRING/~~AUTUMN~~ 2023)

Course (Subject): Computer Vision Code: CE-632

Duration : 03 HOURS Max Marks: 50

Note:

1. Answer **ALL** questions.
2. Marks are indicated against each Question.
3. Question paper consists of: 01 Pages.

**Q1.** **[10 Marks]**

Explain Circular Hough Transform in detail? Can we use it for boundary detection? For which kind of applications, Hough Transform is more suitable?

**Q2.** **[10 Marks]**

What do you mean by Morphological operations? Explain opening and closing Morphological operations in detail. Is Morphological operation same as filtering operation? In what kind of applications, Morphological operations are suitable?

**Q3.** **[10 Marks]**

Write a short note scale-invariant feature transform (SIFT) feature extraction method.

**Q4.** **[10 Marks]**

Explain working of Canny Edge Detection Algorithm. Can it be used to detect corners? If no, which methods are suitable for corner detection?

**Q5.** **[10 Marks]**

Explain HoG (Histogram of Gradients) feature descriptor. How it is calculated? For which applications, it can be used?

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DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(Deemed University)

M.Tech/MS/PhD Examinations

Assessment – I - (Spring 2024)

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Course Name: Deep Learning for Computer Vision | Course Code: AM639

Date: 31 – 01 - 2023

Duration: 01 Hr | Max Marks: 10

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Note:

1. Marks are indicated against each question. Write clearly and legibly Attempt all questions. Calculators are allowed.
2. All programming questions to be answered in Python 3.x

| | | |
|---|---|---|
| Q1 | List and explain the various activation functions used in the modeling of artificial neurons. Also, explain their suitability concerning applications. | [2.5] |
| Q2 | Compare and contrast the single-layered model and multi-layered perceptron model. | [2.5] |
| Q3 | Explain the steps/pseudocode for the perceptron learning algorithm. | [2.5] |
| Q4 | Realize the Boolean inhibition function with the following truth table using a perceptron single-layer/multi-layer model where x and y are inputs. | [2.5] |

| x | y | Output |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

###### ALL THE BEST ######

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(Deemed University)

M.Tech/MS/PhD Examinations

Assessment – II - (Spring 2024)

Course Name: Deep Learning for Computer Vision — Course Code: AM639

Date: 12 – 03 - 2024

Duration: 01 Hr — Max Marks: 10

Note:

1. Marks are indicated against each question. Write clearly and legibly Attempt all questions. Calculators are allowed.

**Q1** Compare Deep Learning with classical Machine learning paradigm. **[2.5]**

**Q2** Consider an image I (4 x 4 x 1) in a CNN given below: **[2.5]**

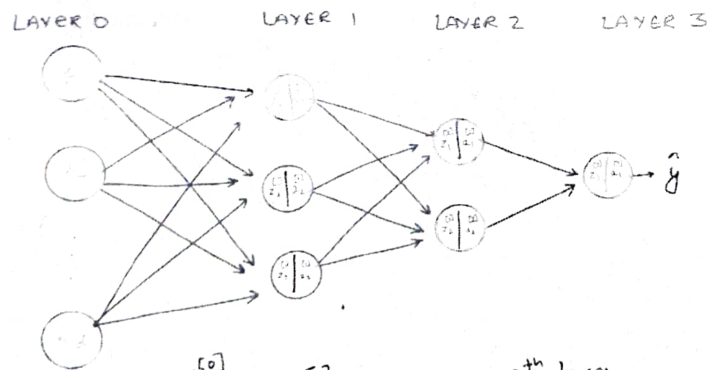
| | | | |
|---|---|---|---|
| 10 | 0 | 3 | 2 |
| 0 | 10 | 5 | 4 |
| 3 | 5 | 10 | 5 |
| 2 | 4 | 5 | 10 |

Find the convolution with a filter K considering stride =1 and zero padding:

| | | |
|---|---|---|
| -1 | 0 | 1 |
| 0 | -1 | 0 |
| 1 | 0 | -1 |

**Q3** It is a well-established fact that the XOR problem cannot be solved by a single layer two input perceptron (not using a hidden layer) since it is not linearly separable. The understanding is that there is no linear function that can separate the classes. However, what if we use a non-monotonic activation function like ReLU, sin() or cos(). is this still the case? Can these types of functions bring in the required nonlinearity to separate them the classes? Support your answers with proof. **[2.5]**

**Q4** Derive the vectorized output of all units for a relu activation in the following network. **[2.5]**




$a^{[0]} \in R^d$ or $R^{N^{[0]}}$ $\quad$ $N^{[0]}$ = # units in $0^{th}$ layer

$d$ – # i/p features

###### **ALL THE BEST** ######

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(Deemed University)

M.Tech/MS/PhD Examinations

Assessment – III - (Spring 2024)

Course Name: Deep Learning for Computer Vision — Course Code: AM639

Date: 16 – 04 - 2024

Duration: 01 Hr — Max Marks: 10

Note:

1. Marks are indicated against each question. Write clearly and legibly Attempt all questions. Calculators are allowed.

| | | |
|---|---|---|
| Q1 | In class, we learned how the ReLU activation function ($ReLU(z) = max(0,z)$) could "die" or become saturated/inactive when the input is negative.<br>(a) A friend of yours suggests the use of another activation function $f(z) = max(0.2z, 2z)$ which he claims will remove the saturation problem. Would this address the problem? Why or why not?.<br>(b) The same friend also proposes another activation function $g(z) = 1.5z$. Would this be a good activation function? Why or why not? | [3] |
| Q2 | What are the limitations of RNN for modelling sequential data? | [2] |
| Q3 | In the class, we saw image captioning as an application of LSTM/GRUs. Assume training data with 100 videos each 10 seconds in duration and captions for each video is available to you.<br>a) Propose a model for captioning a video. Draw a neat sketch of the model along with the state equations at each step wherever necessary.<br>b) Explain the Backpropagation through time for the proposed model. | [5] |

###### **ALL THE BEST** ######

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# DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY
## School of Computer Engineering & Mathematical Sciences
## M.Tech/PhD End Semester- Spring 2024 Examinations

**Roll No:** 23-27-26

**Date:** 30-05-2024 **Total Number of Questions:** **[05]**

**Course Title: Deep Learning for Computer Vision**

**Course Code: AM639**

**Time Allowed: 3.0 Hours** **Max Marks: [50]**

Instructions / NOTE

i. Attempt All Questions.
ii. Support your answer with neat freehand sketches/diagrams, wherever appropriate.
iii. Assume any missing data to suit the case/derivation / answer.
iv. The following calculators or lower models are allowed:
Casio FX-83{ES or GTplus} or FX-85{ES or GTplus}
Sharp Writeview EL-W531, Hewlett Packard HP10S or HP300S , TexasTI-30IIS

| | | |
|---|---|---|
| Q1<br>[10 Marks] | Explain the following terms: | |
| | (a) Leaky ReLU | [02] |
| | (b) Dropout and Max Pooling with example | [02] |
| | (c) Transposed Convolution with example | [02] |
| | (d) Given 128x128x3 and 13 filters of size 8x8x3, what will be the dimension of the output volume when the stride=2 and no padding. | [02] |
| | (e) Consider an image I (4 x 4 x 1) in a CNN given below: | [02] |

| | | | |
|---|---|---|---|
| 10 | 0 | 3 | 2 |
| 0 | 10 | 5 | 4 |
| 3 | 5 | 10 | 5 |
| 2 | 4 | 5 | 10 |

Find the convolution with a filter K considering stride =1 and 'same' padding:

| | | |
|---|---|---|
| -1 | 0 | -1 |
| 0 | -1 | 0 |
| -1 | 0 | -1 |

| | | |
|---|---|---|
| Q2<br>[10 Marks] | (a) Design a perceptron network to solve (using perceptron learning algorithm) a classification problem with four classes of input vector. The four classes are: | [05] |

$$\text{class 1: } \left\{ \mathbf{p}_1 = \begin{bmatrix}1\\1\end{bmatrix}, \mathbf{p}_2 = \begin{bmatrix}1\\2\end{bmatrix} \right\}, \text{ class 2: } \left\{ \mathbf{p}_3 = \begin{bmatrix}2\\-1\end{bmatrix}, \mathbf{p}_4 = \begin{bmatrix}2\\0\end{bmatrix} \right\},$$

$$\text{class 3: } \left\{ \mathbf{p}_5 = \begin{bmatrix}-1\\2\end{bmatrix}, \mathbf{p}_6 = \begin{bmatrix}-2\\1\end{bmatrix} \right\}, \text{ class 4: } \left\{ \mathbf{p}_7 = \begin{bmatrix}-1\\-1\end{bmatrix}, \mathbf{p}_8 = \begin{bmatrix}-2\\-2\end{bmatrix} \right\}.$$

| | | |
|---|---|---|
| | (b) Design and propose a model for image captioning using an encoder-decoder network by deriving all forward-pass equations of the network. Draw a neat sketch of the model along with the state equations at each step wherever necessary. For an input image of unknown size, the model should generate a caption. Explain all encoding considered for the solution. | [05] |

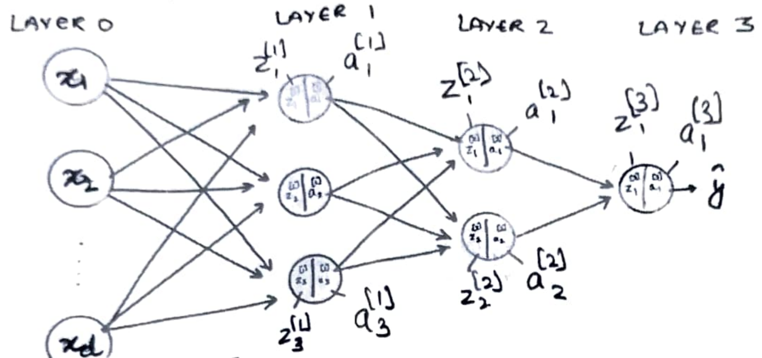
**Q3** [10 Marks]

(a) Explain in detail Backpropagation Through Time (BPTT) for training a recurrent neural network (RNN). Derive all necessary gradients w.r.t. loss function. [05]

(b)Explain Selective Read, Selective Write, and Selective forget in LSTM using suitable state equations. [05]

**Q4** [10 Marks]

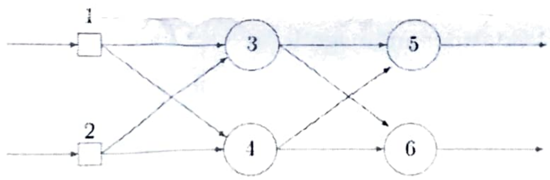
(a) For a classification problem and sigmoid activation in the hidden layers. Compute $\frac{\partial L}{\partial z^{[1]}}$ [05]




$a^{[0]} \in R^{d}$ or $R^{N^{[0]}}$ $\quad N^{[0]}$ = # Units in $0^{th}$ Layer

$d \rightarrow$ # i/p features

(b) The following diagram represents a feed-forward neural network with one hidden layer: [05]




A weight on connection between nodes *i* and *j* is denoted by $w_{ij}$, such as $w_{13}$ is the weight on the connection between nodes 1 and 3. The following table lists all the weights in the network:

| | |
|---|---|
| $w_{13} = -2$ | $w_{35} = 1$ |
| $w_{23} = 3$ | $w_{45} = -1$ |
| $w_{14} = 4$ | $w_{36} = -1$ |
| $w_{24} = -1$ | $w_{46} = 1$ |

$$\varphi(v) = \begin{cases} 1 & \text{if } v \geq 0 \\ 0 & \text{otherwise} \end{cases} \qquad \varphi(v)$$

Each of the nodes 3, 4, 5 and 6 uses the activation function where $v$ denotes the weighted sum of a node. Each of the input nodes (1 and 2) can only receive binary values (either 0 or 1). Calculate the output of the network ($y_5$ and $y_6$) for each of the input patterns:

| Pattern: | $P_1$ | $P_2$ | $P_3$ | $P_4$ |
|---|---|---|---|---|
| Node 1: | 0 | 1 | 0 | 1 |
| Node 2: | 0 | 0 | 1 | 1 |

| | | |
|---|---|---|
| Q5<br>[10 Marks] | (a) Propose a GAN-inspired image super-resolution architecture for enhancing the resolution of the images captured by surveillance drones. Derive the loss function using the binary cross entropy loss function for the proposed model. You may assume a loss function, or nonlinearity suitable for this task.<br>(b) The loss function of GANs has two components: the loss function for the generator and the discriminator. Explain mathematically why the discriminator seeks to maximize the given quantity and the generator seeks to achieve the reverse. | [07]<br><br><br><br>[03] |

###### ALL THE BEST ######